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# Combustion characteristics of alternative gaseous fuels

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**Abstract**

Fundamental flame properties of mixtures of air with hydrogen, carbon monoxide, and $C_1$–$C_4$ saturated hydrocarbons were studied both experimentally and numerically. The fuel mixtures were chosen in order to simulate alternative gaseous fuels and to gain insight into potential kinetic couplings during the oxidation of fuel mixtures. The studies included the use of the counterflow configuration for the determination of laminar flame speeds, as well as extinction and ignition limits of premixed flames. The experiments were modeled using the USC Mech II kinetic model. It was determined that when hydrocarbons are added to hydrogen flames as additives, flame ignition, propagation, and extinction are affected in a counterintuitive manner. More specifically, it was found that by substituting methane by propane or *n*-butane in hydrogen flames, the reactivity of the mixture is reduced both under pre-ignition and vigorous burning conditions. This behavior stems from the fact that propane and *n*-butane produce higher amounts of methyl radicals that can readily recombine with atomic hydrogen and reduce thus the rate of the $H + O_2 \rightarrow O + OH$ branching reaction. The kinetic model predicts closely the experimental data for flame propagation and extinction for various fuel mixtures and pressures, and for various amounts of carbon dioxide in the fuel blend. On the other hand, it underpredicts, in general, the ignition temperatures.
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*Keywords:* Alternative fuels; Flame propagation; Flame extinction; Flame ignition; Pressure effects

## 1. Introduction

Current trends in fuel prices, world politics, and environmental regulations have renewed interest in alternative sources of energy. Renewable and short carbon-cycle fuels such as gasified coal, waste- and biomass-derived gas, and by-product gases from coke manufacturing are among the viable alternatives. Such fuels are typically mixtures of hydrogen ($H_2$), carbon monoxide (CO), and $C_1$–$C_4$ hydrocarbons, and frequently notable quantities of carbon dioxide ($CO_2$), which is a highly efficient chaperon molecule in three-body radical recombination reactions.

Flame properties of the neat fuels mentioned above, have been studied extensively both experimentally (e.g. [1–3]) and numerically (e.g. [4–7]). Additionally, several kinetic models have been published for $H_2$, CO, and $C_1$–$C_4$ hydrocarbons (e.g. [2,8–10]) and have been validated largely against experiments involving neat components (e.g. [11–13]) or $H_2$/CO mixtures (e.g. [14]). Potential kinetic couplings stemming from fuel blending have been assessed to lesser extent, with the main exception being the addition of $H_2$ to hydrocarbons (e.g. [15,16]) especially to $CH_4$ (e.g. [17–19]). However, kinetic interactions between $H_2$ and CO with $CH_4$, as well as fuel blends involving

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propane ($C_3H_8$) or $n$-butane ($n$-$C_4H_{10}$) have not been studied systematically.

Based on those considerations, the goal of this study was to provide archival experimental flame data for blends of $H_2$, CO, $CH_4$, $C_3H_8$, and $n$-$C_4H_{10}$ and to provide insight into the underlying kinetic couplings. The study was conducted for premixed flames in the counterflow configuration, and the combustion characteristics in focus were laminar flame speeds, $S_u^\circ$; extinction strain rates, $K_{ext}$; and ignition temperatures, $T_{ign}$. It is of interest to note that flame ignition in well-controlled environments, such as stagnation flows, has been studied mainly for non-premixed flames (e.g. [20–22]) and there are only a few studies [23,24] on the ignition of premixed flames.

## 2. Experimental approach

The experiments were carried out in the opposed jet configuration (e.g. [15,25–27]). The majority of the measurements were performed for fuel lean mixtures due to their relevance to lean-burning applications. The propagation and extinction studies were performed in a variable-pressure chamber for pressures ranging from $p = 1$ to 4 atm and for unburned mixture temperature $T_u = 298$ K. The burner diameters were 9 and 14 mm.

$S_u^\circ$'s were determined using the twin flame configuration. Two symmetrical and planar flames were established and the axial velocity profile along the stagnation streamline of the flow was measured using digital particle image velocimetry. The flow was seeded with 0.3 μm diameter silicon oil droplets produced by a high efficiency nebulizer. The minimum axial flow velocity upstream of the flame is defined as the reference flame speed, $S_{u,ref}$ and the maximum absolute value of the velocity gradient upstream of the flame is defined as the strain rate, $K$, (e.g. [15,25–27]). $S_u^\circ$ was determined through non-linear extrapolation of $S_{u,ref}$ to $K = 0$ using a computationally-assisted approach developed recently by the authors [26,27].

$K_{ext}$'s were measured using the single-flame configuration resulting from counterflowing fuel/air mixtures against an ambient temperature $N_2$ jet. The flow rate of the ambient temperature $N_2$ was determined such that the momentum fluxes from the bottom and top jet are equal. A flame was established at a near extinction condition and the oxidizer flow rate was slightly modified to achieve extinction and determine thus $K_{ext}$ (e.g. [26,27]). The burner diameters used were $D = 14$ mm for $K_{ext} \leqslant 500\ s^{-1}$ and $D = 9$ mm for $K_{ext} > 500\ s^{-1}$. The burner separation distance, $L$ was kept equal to $D$, i.e., $L/D = 1$.

$T_{ign}$ was measured by counterflowing fuel/air mixtures at $T_u = 423$ K against a heated $N_2$ flow exiting from a quartz tube in which a silicon carbide spiral-heating element is embedded and capable of heating the $N_2$ up to 1400 K (e.g. [20,21,24]). The fuel/air jet was heated using a ceramic heating jacket that maintained $T_u = 423$ K at the lower burner exit. The inner diameters of both burner nozzles and the nozzle separation distance were 22 mm. Ignition was achieved by establishing the appropriate flow rates and temperatures at the two burner exits, and by increasing subsequently the fuel flow rate until ignition was observed. The temperature of the $N_2$ jet at the upper burner exit, $T_{N_2,\ exit}$, was measured with a R-type thermocouple and corrected for radiation/convective heat transfer (e.g. [20,21,24]). The reported $T_{ign}$ is defined as the measured $T_{N_2,\ exit}$ at the state of ignition.

The fuel blends tested in this study are outlined in Table 1. Fuels-1 and -2 simulate products of the destructive distillation of coal, otherwise referred to as by-product coke oven gas and contain large amounts of $H_2$ [28]. Fuels-3 and -4 represent synthesis gas, which is produced via steam reforming coal or other condense-phase hydrocarbon fuels, and it contains largely CO and $H_2$, two particularly clean-burning fuels [29]. Fuels-5 and -6 are surrogate mixtures that simulate digester and landfill gas, a fuel produced by the biological reduction of organic solids and they consist primarily of methane and carbon dioxide [30,31]. The $2\sigma$ standard deviations in $S_u^\circ$'s are indicated in all figures. The uncertainty in equivalence ratio, $\phi$, was less than 0.5% and in $T_{ign}$ of the order of ±20 °C.

## 3. Numerical approach

$S_u^\circ$'s were computed using the PREMIX code [32,33]. The counterflow configuration was simulated using the code of Kee and coworkers [34], which has been modified to allow for the simulation of asymmetric boundary conditions [35,36] and to account for thermal radiation from $CH_4$, CO, $CO_2$, and $H_2O$ at the optically thin limit

Table 1
Composition of fuel blends on molar basis.

| Fuel | Blended fuel composition |
|---|---|
| 1 | 60% $H_2$ + 10% CO + 30% $CH_4$ |
| 1a | 60% $H_2$ + 10% CO + 30% $C_3H_8$ |
| 1b | 60% $H_2$ + 10% CO + 30% $n$-$C_4H_{10}$ |
| 2 | 95% $H_2$ + 5% $CH_4$ |
| 2a | 95% $H_2$ + 5% $C_3H_8$ |
| 2b | 95% $H_2$ + 5% $n$-$C_4H_{10}$ |
| 3 | 32% $H_2$ + 58% CO + 10% $CH_4$ |
| 3a | 32% $H_2$ + 58% CO + 10% $C_3H_8$ |
| 4 | 54% $H_2$ + 11% CO + 25% $CH_4$ + 10% $CO_2$ |
| 5 | 55% $CH_4$ + 45% $CO_2$ |
| 6 | 75% $CH_4$ + 25% $CO_2$ |

[37]. Both codes are integrated with the CHEMKIN [38] and the Sandia Transport [39] subroutine libraries. The H and $H_2$ diffusion coefficients of several key pairs are based on a recently updated set of Lennard–Jones parameters [2]. Both the mixture-averaged and full multi-component transport coefficient formulations were used in the simulations along with the Soret effect.

$K_{ext}$'s were computed by establishing a vigorously burning flame at a given $K$, which was increased subsequently until a turning-point behavior was realized, indicative of extinction. In a recent study by the authors [27], the axial velocity gradient along the centerline at the burner exit, denoted as $\alpha$, was found to have a considerable effect on the numerical determination of $K_{ext}$. Thus, in computing $K_{ext}$, the experimental values of $\alpha$ were used as an additional boundary condition. Similarly, $T_{ign}$'s were computed by establishing first a cold non-reacting solution for a low $T_{N_2,\ exit}$, and by subsequently increasing it until a turning-point behavior was realized, indicative of ignition. For both cases, solution around the turning-points was obtained imposing a value for a radical species concentration at the location of the maximum slope of the profile, and solving for the velocities at the burner exits, which determine the magnitude of $K$ or $T_{N_2,\ exit}$ [40,41]. Thus, $K$ or $T_{N_2,\ exit}$ becomes a dependent variable.

The effects of chemical kinetics and molecular diffusion on $S_u^\circ$, $K_{ext}$, and $T_{ign}$ were assessed through sensitivity analysis. While the standard CHEMKIN-based codes allow for automated sensitivity analysis with respect to rate constants for $S_u^\circ$ as well as temperature and species concentrations, this is not the case for $K_{ext}$ and $T_{ign}$. In this investigation, this was achieved by invoking the aforementioned continuation approach that renders $K$ or $T_{N_2,\ exit}$ dependent variables, so that their sensitivity on kinetics and transport can be computed from first principle.

The USC Mech II kinetic model [42] that consists of 111 species and 784 elementary reactions was used in the simulations without any modifications.

## 4. Results and discussion

Figure 1a depicts experimental and computed $S_u^\circ$'s of flames of Fuel-1 (60% $H_2$ + 10% CO + 30% $CH_4$) and Fuel-3 (32% $H_2$ + 58% CO + 10% $CH_4$) at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$. While the computed $S_u^\circ$'s of Fuel-3/air flames are in good agreement with the experimental ones, $S_u^\circ$'s of Fuel-1/air flames are under predicted by the simulations. Over the $\phi$-range considered, Fuel-1/air and Fuel-3/air flames have similar $S_u^\circ$'s even though Fuel-1 contains approximately twice the amount of $H_2$ compared to Fuel-3. This is primarily due to the fact that the adiabatic flame temperature, $T_{ad}$, of Fuel-3/air flames is approximately 100 K higher than $T_{ad}$, of Fuel-1/air flames. Additionally, there is less than half the amount of $CH_4$ in Fuel-3/air flames compared to Fuel-1/air flames.

The experimental and computed $S_u^\circ$'s of flames of Fuels-1 and -4 (54% $H_2$ + 11% CO + 25% $CH_4$ + 10% $CO_2$) at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$ are compared in Fig. 1b. The composition of Fuel-4 is similar to Fuel-1 with the exception of the addition of $CO_2$ to Fuel-4 and as a result $S_u^\circ$'s of Fuel-4/air flames are lower. The computed $S_u^\circ$'s of Fuel-4/air flames are in agreement with the experimental data.

Figure 2a depicts the experimental and computed $K_{ext}$'s of Fuel-1/air and Fuel-3/air flames at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$. Similarly to $S_u^\circ$, the experimental $K_{ext}$'s of Fuel-1/air and Fuel-3/air flames are very close to each other. On the other hand, the model underpredicts the experimental data for both Fuel-1/air and Fuel-3/air flames as $\phi$ increases.

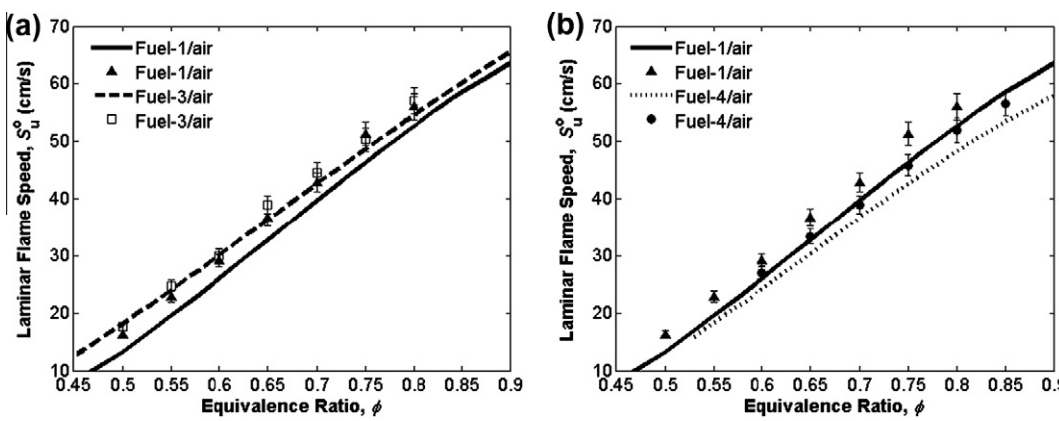


Fig. 1. Experimental and computed $S_u^\circ$'s of (a) Fuel-1/air and Fuel-3/air flames and (b) Fuel-1/air and Fuel-4/air flames at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$. Symbols: experimental data; Lines: numerical results. The uncertainty bars indicate $2\sigma$ standard deviations.

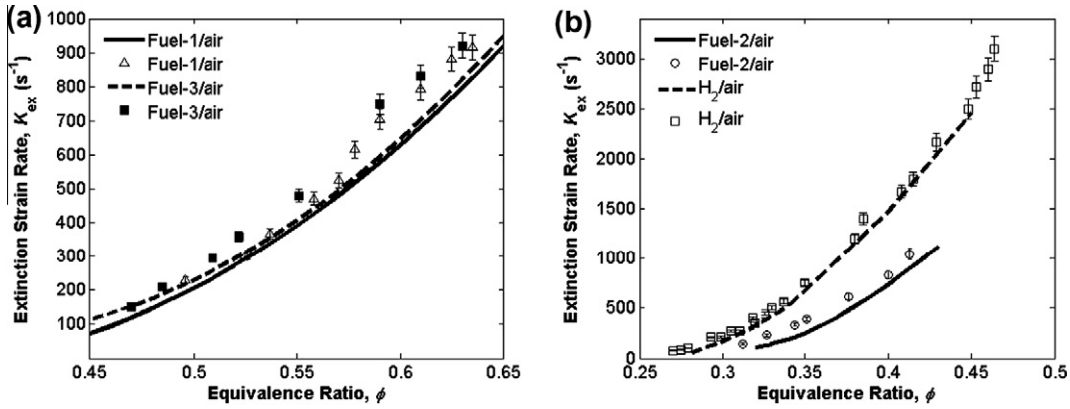


Fig. 2. Experimental and computed $K_{ext}$'s of (a) Fuel-1/air and Fuel-3/air flames and (b) $H_2$/air and Fuel-2/air flames at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$. Symbols: experimental data; Lines: numerical results. The uncertainty bars indicate $2\sigma$ standard deviations.

In Fig. 2b, the experimental and computed $K_{ext}$'s of $H_2$/air and Fuel-2 (95% $H_2$ + 5% $CH_4$)/air flames at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$ are compared. As expected, the addition of trace amounts of $CH_4$ to $H_2$/air flames reduces $K_{ext}$. The model predicts closely both sets of experimental data.

Additional experiments were carried out, by substituting $CH_4$ in Fuel-1 with equal amounts $C_3H_8$ (Fuel-1a) and $n$-$C_4H_{10}$ (Fuel-1b) on molar basis, while the $H_2$ and CO mole fractions were kept equal to that in Fuel-1. Both experimental and computed $S_u^\circ$'s of Fuel-1a/air and Fuel-1b/air flames are lower compared to Fuel-1/air at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$, as depicted in Fig. 3a.

Similarly, the 5% $CH_4$ in Fuel-2 was substituted with 5% $C_3H_8$ (Fuel-2a). In Fig. 3b, $S_u^\circ$'s of Fuels-2 and -2a at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$ are shown. Consistent with observations in Fig. 3a, the substitution of $CH_4$ with $C_3H_8$ results in a reduction of $S_u^\circ$.

Since the numerical simulations captured the effect of substituting $CH_4$ by a heavier fuel compound, the computed flame structures were used to provide insight into the controlling mechanisms. Additionally, it was determined that the substitution of $CH_4$ by $C_3H_8$ or $n$-$C_4H_{10}$, affects $T_{ad}$ at most by 10 K for Fuel-1 and by negligible amounts for Fuel-2, given that in both cases the hydrocarbons are additives. Thus, the observed effect is not of thermal but rather of kinetic nature.

Figure 4a depicts the species profiles of methyl ($CH_3$) and hydrogen (H) radicals for $\phi = 0.6$ Fuel-1/air, Fuel-1a/air, and Fuel-1b/air flames at $p = 1$ atm and $T_u = 298$ K. Similar species profiles for $\phi = 0.6$ Fuel-2/air, Fuel-2a/air, and Fuel-2b/air flames are shown in Fig. 4b. For both Fuels-1 and -2, substituting $CH_4$ by either $C_3H_8$ or

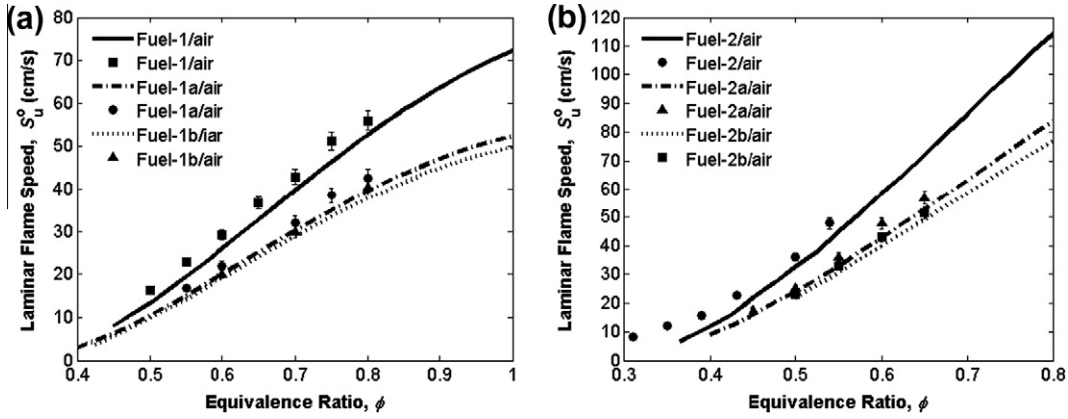


Fig. 3. Experimental and computed $S_u^\circ$'s of (a) Fuel-1/air, Fuel-1a/air, and Fuel-1b/air flames and (b) Fuel-2/air, Fuel-2a/air, and Fuel-2b/air flames at $p = 1$ atm, $T_u = 298$ K, and $\phi < 1.0$. Symbols: experimental data; Lines: numerical results. The uncertainty bars indicate $2\sigma$ standard deviations.

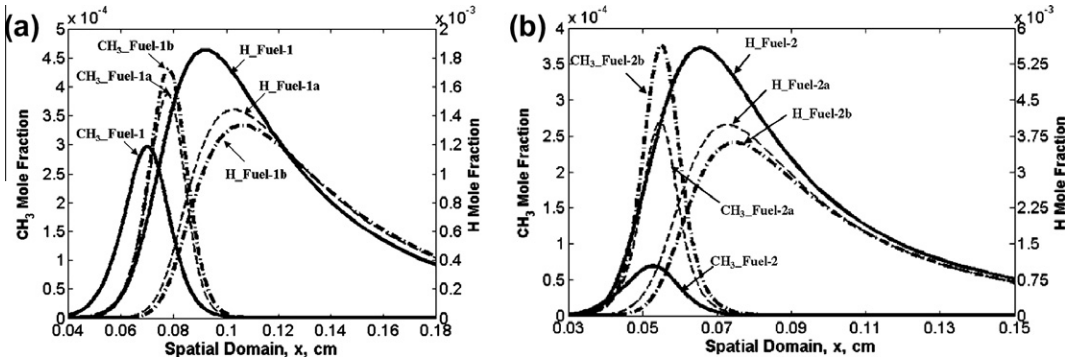


Fig. 4. Computed $CH_3$ and H mole fraction profiles for $\phi = 0.6$ (a) Fuel-1/air, Fuel-1a/air, and Fuel-1b/air flames and (b) Fuel-2/air, Fuel-2a/air, and Fuel-2b/air flames at $p = 1$ atm and $T_u = 298$ K.

$n$-$C_4H_{10}$, results in a notable increase of $CH_3$ concentration and a corresponding decrease of H concentration.

In Fuel-1a/air and Fuel-2a/air flames, ethylene ($C_2H_4$) forms from the thermal decomposition of $n$-propyl ($n$-$C_3H_7$)

$$n\text{-}C_3H_7 \rightarrow C_2H_4 + CH_3. \quad \text{(R1)}$$

Subsequently, $C_2H_4$ reacts with H to form $CH_3$ via the following sequence

$$C_2H_4 + H + M \rightarrow C_2H_5 + M, \quad \text{(R2)}$$

$$C_2H_5 + H \rightarrow CH_3 + CH_3. \quad \text{(R3)}$$

In Fuel-1b/air and Fuel-2b/air flames, $C_2H_5$ forms from the thermal decomposition of $p$-butyl ($p$-$C_4H_9$)

$$p\text{-}C_4H_9 \rightarrow C_2H_4 + C_2H_5 \quad \text{(R4)}$$

and $CH_3$ is produced via R3.

Reaction path analysis of Fuel-2a/air flames reveals that $CH_3$ is additionally produced from acetaldehyde ($CH_3CHO$) via

$$i\text{-}C_3H_7 + O \rightarrow CH_3CHO + CH_3, \quad \text{(R5)}$$

$$CH_3CHO + X \rightarrow CH_3CO + XH, \quad \text{(R6)}$$

$$CH_3CO \rightarrow CH_3 + CO. \quad \text{(R7)}$$

where X denotes H, O, or $OH^{\bullet}$ $CH_3$ then scavenges H through

$$CH_3 + H + M \rightarrow CH_4 + M. \quad \text{(R8)}$$

When $CH_4$ is substituted by $n$-$C_4H_{10}$ (Fuels-1b and -2b), notable amounts of $CH_3CHO$ are produced from reactions of $C_2H_5$ and $s$-butyl ($s$-$C_4H_9$)

$$C_2H_5 + O \rightarrow CH_3CHO + H \quad \text{(R9)}$$

$$s\text{-}C_4H_9 + HO_2 \rightarrow CH_3CHO + C_2H_5 + OH \quad \text{(R10)}$$

and from 2-butene (2-$C_4H_8$) via

$$2\text{-}C_4H_8 + O \rightarrow CH_3CHO + C_2H_4 \quad \text{(R11)}$$

resulting again in higher concentrations of $CH_3$ via the R6 and R7 sequence.

In summary, adding small amounts of $C_3H_8$ or $n$-$C_4H_{10}$ to $H_2$/air flames results in increased consumption of H through reactions with $C_2H_4$ and $C_2H_5$, and increased concentrations of $CH_3$ that in turn consumes H, lowering thus the rate of the main branching reaction

$$H + O_2 \rightarrow O + OH \quad \text{(R12)}$$

and the attendant values of $S_u^{\circ}$ and $K_{ext}$. Sensitivity analysis and reaction path analysis are presented in Figs. S1–S3 in the Supplementary data.

Figure 5a depicts the experimental and computed $T_{ign}$'s of Fuel-1/air, Fuel-2/air, and Fuel-3/air mixtures at $p = 1$ atm, $T_u = 423$ K, $K = 200\ s^{-1}$, and $\phi < 1.0$. $T_{ign}$'s are lowest for Fuel-2/air and highest for Fuel-1/air mixtures. This is expected since Fuel-2 has the highest concentration of $H_2$ of all three mixtures. For all three fuel/air mixtures, $T_{ign}$ is relatively insensitive to $\phi$.

The computed $T_{ign}$'s, also shown in Fig. 5a, are in good agreement with the experimental data for Fuel-2/air and Fuel-3/air mixtures, and they are slightly lower than the experimental data for Fuel-1/air mixtures. Sensitivity analysis of the $T_{ign}$ to kinetics indicates that $T_{ign}$ is sensitive mainly to the branching reaction R12 and the termination reaction

$$H + O_2 + M \rightarrow HO_2 + M. \quad \text{(R13)}$$

Figure 5b depicts the experimental and computed $T_{ign}$'s of Fuel-3/air and Fuel-3a/air mixtures at $p = 1$ atm, $T_u = 423$ K, $K = 200\ s^{-1}$, and $\phi < 1.0$. Both experiments and simulations indicate that $T_{ign}$'s of Fuel-3a/air are higher compared to Fuel-3/air mixtures by 30–35 K. This suggests that the substitution of $CH_4$ by $C_3H_8$ reduces the ignition propensity of the mixture, a result that is consistent with the attendant effect on flame propagation and extinction.

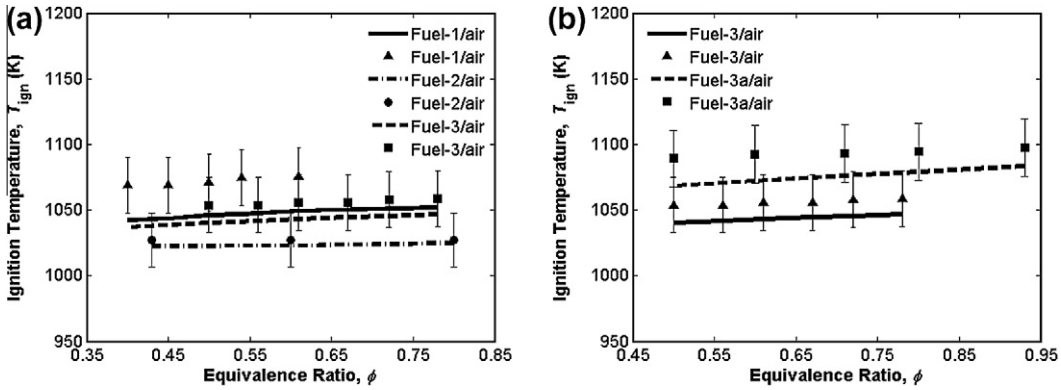


Fig. 5. Experimental and computed $T_{ign}$'s of (a) Fuel-1/air, Fuel-2/air, and Fuel-3/air mixtures and (b) Fuel-3/air and Fuel-3a/air mixtures at $p = 1$ atm, $T_u = 423$ K, $K = 200\ s^{-1}$, and $\phi < 1.0$. The uncertainty bars indicate $2\sigma$ standard deviations.

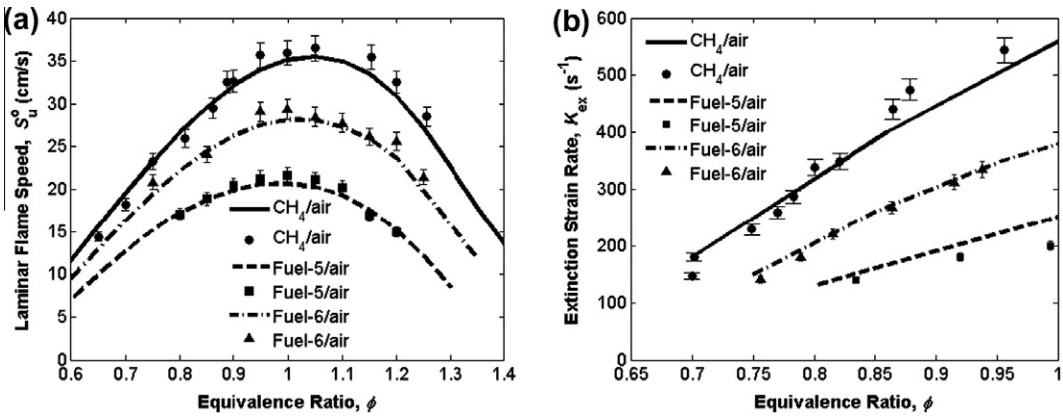


Fig. 6. Experimental and computed (a) $S_u^\circ$'s and (b) $K_{ext}$'s of $CH_4$/air, Fuel-5/air, and Fuel-6/air flames at $p = 1$ atm and $T_u = 298$ K. The uncertainty bars indicate $2\sigma$ standard deviations.

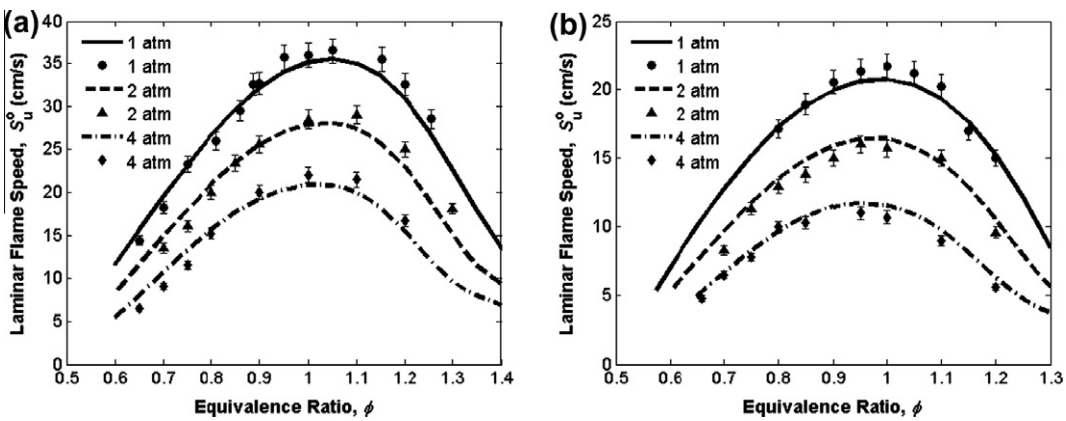


Fig. 7. Experimental and computed $S_u^\circ$'s of (a) $CH_4$/air flames (b) Fuel-5/air flames at $T_u = 298$ K and $p = 1$, 2, and 4 atm. The uncertainty bars indicate $2\sigma$ standard deviations.

The effect of $CO_2$ on flame propagation and extinction was assessed, and Fig. 6a and b depict the experimental and computed $S_u^\circ$'s and $K_{ext}$'s, respectively for $CH_4$/air, Fuel-5 (55% $CH_4$ + 45% $CO_2$)/air, and Fuel-6 (75% $CH_4$ + 25% $CO_2$)/air flames at $p = 1$ atm and $T_u = 298$ K. As expected, Fuel-5/air flames have the lowest $S_u^\circ$'s and $K_{ext}$'s. In addition, the peak of $S_u^\circ$ shifts closer to $\phi = 1.0$ as the $CO_2$ concentration in the fuel increases because $T_{ad}$ is lower and product dissociation is suppressed. The simulations capture the effect of $CO_2$ accurately.

The pressure effect on $S_u^\circ$ was assessed also for $CH_4$/air and Fuel-5/air flames, by performing experiments at $p = 1$, 2, and 4 atm and $T_u = 298$ K, and the results are shown in Fig. 7a and b. As expected, $S_u^\circ$ is reduced with pressure for all $\phi$'s as $S_u^\circ \sim p^{n/2-1}$, and the overall reaction order, $n$, is less than 2 (e.g. [43]). The computed $S_u^\circ$'s are in good agreement with the current experimental data over all $\phi$'s considered and for the pressure range of 1–4 atm. Previously published $CH_4$/air $S_u^\circ$ data can be found in Refs. [1,44].

## 5. Concluding remarks

Laminar flame speeds, as well as ignition and extinction limits of mixtures of hydrogen, carbon monoxide, and $C_1$–$C_4$ saturated hydrocarbons with air were studied experimentally and numerically in the counterflow configuration. The fuel mixtures were chosen to simulate alternative gaseous fuels. Laminar flame speeds were determined at ambient unburned gas temperature and pressures ranging from one to four atmospheres, while extinction limits were determined at ambient unburned gas temperature and pressure. Ignition temperatures were determined by counterflowing heated nitrogen against heated fuel/air mixtures at atmospheric pressure. The experimental flame data of these mixtures are of practical important and they also provide information that can be used to assess the effects of kinetic couplings during the oxidation of fuel mixtures involving hydrogen, carbon monoxide, methane, propane, and $n$-butane. The experiments were modeled using detailed descriptions of chemical kinetics and molecular transport. The experimental results revealed that the substitution of methane in a fuel mixture with either propane or $n$-butane results in the reduction of the mixture's reactivity both under pre-ignition and vigorous burning conditions. The numerical simulations reproduced closely the experimental laminar flame speeds, extinction strain rates, and ignition temperatures over all experimental conditions.

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## Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.116.

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